Free-Tier LLM Gateways as AI-Text Detectors

A dated benchmark of model accuracy, quota regimes and out-of-sample stability across four providers, with a band-scoring protocol for production use

Preprint / technical report — not peer reviewed.
Measurement window: 2026-09-28 to 2026-10-13. All availability and quota claims are time-bound to this window.
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Abstract. General-purpose LLMs asked to output a structured AI-ness verdict can serve as text detectors at near-zero cost, and their availability is governed less by model quality than by the free-tier policies of the gateways that serve them. This report benchmarks 20+ models across four gateways (OpenRouter free pool, NVIDIA NIM free tier, Google AI Studio free tier, and a local multi-account router) on a 48-text in-sample corpus and a 48-text disjoint holdout drawn from HC3, all scored through the same strict-JSON prompt, a fixed decision band (AI ≥ 0.69, HUMAN ≤ 0.40) and a confidence gate (≥ 0.60). Three findings stand out. First, *model choice dominates tuning*: identical prompts and parsers span 0.31 to 1.00 corpus accuracy, and one free model (`moonshotai/kimi-k3`) was the only configuration measured with zero wrong decisions on both corpora, reproduced in a same-day repeat run (AUC 1.000 both captures; McNemar vs. incumbent lanes p = 1.0000). Second, each gateway enforces a materially different regime — OpenRouter has two 429 classes that require opposite remedies, Google's per-model daily quota (nominally 20) *delivers* a volume that is per model *and* per project (5–46 requests in one window observed) and stacks linearly with extra project keys, and NVIDIA free serving windows open and close within hours — so a free model's measured accuracy and its usability as a live lane are separate, per-gateway questions. Third, in-sample perfection does not transfer: a lane that erred zero times on the corpus its band was fitted on produced a false negative out of sample, while the one lane that reproduced zero-wrong out of sample deferred 2–3 of 48 texts per run. A practitioner's decision table and an explicit limitations section accompany the results.

1. Introduction

Commercial AI-text detectors are calibrated to the models that existed when they were trained, and they are expensive per request. An alternative used here: ask a general-purpose LLM, through a plain chat API, to score a text and emit a structured verdict, then apply fixed decision thresholds. Because the cheapest LLM access today is free-tier gateway capacity, the practical question for anyone building a low-cost verifier splits in two: *which models score well*, and *which*

serving channels stay reachable. This report measures both halves on the same protocol, dated, with every availability claim tied to the gateway and window where it was observed.

The work was performed as engineering validation of a web verification app, and this report is written to be usable as its published evidence base: §2 defines the protocol, §3–§5 give results, §6 is a decision table for choosing a free detector model, §7 states the limitations without softening them.

2. Method

2.1 Corpora

Both corpora are drawn from the Human-vs-AI Generated Comparison corpus (HC3) [1], which pairs real user posts with ChatGPT-3.5-era answers across six domains (medical Q&A, Reddit ELI5, open QA, wikiCSAI, finance, law).

- **In-sample corpus** — 48 texts (24 AI / 24 HUMAN), ~85-word human median.
- **Holdout corpus** — a disjoint 48-text two-sided set (24 AI / 24 HUMAN) taken from a non-overlapping offset window of the same source; disjointness was verified by normalized text overlap and by question overlap (0 of 48 shared), and independently re-checked against the source with a negative control planted on a tampered copy.

A third control corpus of 46 real book excerpts (Project Gutenberg, all-human) was used to test false-positive behaviour and is reported only as a caution: it is one-sided and saturated every model at specificity 1.000, so it cannot rank detectors (see §7).

2.2 Scoring protocol

Every model received the same single-turn prompt requesting a strict JSON object with `ai_probability` (0–1), `confidence` (0–1) and short signals, at temperature 0.1. Notable protocol choices, each learned from failure:

- **No `response_format`.** On free tiers, JSON-mode requests caused HTTP 400s, reasoning preamble leakage and malformed output; it was removed and replaced by a defensive parser (markdown-fence strip → balanced-brace scan → regex salvage). Salvaged rows are always flagged and excluded from headline numbers.
- **Transport failures are separated from model failures.** Each capture records no-answer vs bad-JSON vs HTTP-error vs timeout per row; a coverage gap attributable to a gateway is never quoted as model evidence without its *n*.
- **Every "unreachable" finding names the gateway and date.** The same weights proved reachable on one provider and permanently dead on another (§5).

2.3 Decision band and metrics

Verdicts follow a fixed band: **AI if p ≥ 0.69**, **HUMAN if p ≤ 0.40**, otherwise deferred (review); a deferral also results when the lane reports confidence < 0.60. Metrics:

- `acc/corpus` = correct ÷ texts given (deferral costs accuracy) — the deployment-relevant column;
- `acc/answered` = correct ÷ texts the model actually decided — the judgement-quality column;
- AUC via Mann-Whitney over answered rows (ties 0.5) — ranking quality, threshold-free;
- Paired significance via exact McNemar on decisive-vs-decisive disagreements [2], joining the two captures by text id and applying one rule to both sides.

The band was calibrated on the in-sample corpus, which makes the holdout runs the only unbiased numbers in this report; per-model bands (fitted cuts, two-fold) were also computed for several lanes and are reported where they change the story.

2.4 Gateways under test

Gateway	Access path	Class
OpenRouter	15 `:free` model ids; up to 6 free accounts (class-(b) headroom stacks linearly across accounts)	Free
NVIDIA NIM	free-tier endpoints; 80-model catalogue, ~20 genuinely servable on one account	Free
Google AI Studio	2 working free accounts + 1 rejected at auth (per-model buckets are per project, so each working account adds capacity), ~62 model ids	Free
Local multi-account router	a self-hosted OpenAI-compatible router holding OAuth and key-based upstream accounts	Free + paid
Paid reference pair	`nemotron-3-ultra-550b` primary + `diffusiongemma-26b` tie-break	Paid baseline

3. In-sample results (48-text corpus)

All rows answered 48/48 unless *n* says otherwise; band as in §2.3. Two accuracy columns are given because they answer different questions (see §2.3); only `acc/corpus` is comparable across coverage.

Model	Gateway	n	AUC	wrong	deferred	acc/corpus	med AI	med HUMAN
kimi-k3	NVIDIA free	48	1.000	0	3	0.938	0.860	0.220

glm-5.3	NVIDIA free	48	1.000	0	3	0.938	0.930	0.135
glm-5.3-flash	NVIDIA free	48	1.000	1	2	0.938	0.930	0.210
claude-sonnet-4.5 (router/OAuth)	router	48	0.988	2	0	0.958	0.850	0.150
claude-sonnet-4.5 no-think (router)	router	48	0.986	2	0	0.958	0.850	0.150
ling-3.1-flash	OpenRouter free	48	0.986	1	12	0.729	0.850	0.325
gemini-3.1-flash-lite	AI Studio free	46	0.955	13	0	0.688	0.920	0.835
gemma-4-31b-it	NVIDIA free	48	0.945	7	0	0.854	0.980	0.150
paid 2-model pair (tie-break replay)	NVIDIA paid	48	—	3	0	0.938	—	—
nemotron-3-ultra-550b (paid primary alone)	NVIDIA paid	48	—	2	10	0.750	0.815	0.250
openrouter/free (router model)	OpenRouter free	33	0.743	7	7	0.396	0.850	0.680
lfm-2.5-2.6b:free	OpenRouter free	47	0.480	16	16	0.313	0.665	0.650
llama-3.2-11b-vision-instruct	NVIDIA free	48	≈0.0	32	0	0.333	0.200	0.700

Observations that survived the whole measurement window:

- **Model choice dominates tuning.** The same prompt and parser span 0.313 → 0.938 acc/corpus; temperature/top_p sweeps move a model far less than switching models does.
- **AUC cannot see a band problem.** Several lanes ranked classes perfectly while flagging most humans as AI under the shipped cuts (a router route at AUC 1.000 with human median 0.72; a Google lite lane at AUC 0.955 with human median 0.835 and 13 of 24 humans misclassified, §5.2). Only band-scoring exposes this; ranking alone is not evidence of deployability.

- **Detectors can be inverted.** A free vision model scored human text as *more* AI-like than AI text (separation −0.500). Any unmeasured model is a liability; this is the case for labelling every detector choice out of sample.
- **Run-to-run jitter is a pipeline property.** Three runs of one identical model scored 0.458 / 0.404 / 0.378 on the same corpus; on a borderline screen subset, 8 of 12 texts flipped verdicts between identical reruns. Single-text or single-run judgements are meaningless; aggregates and paired tests are the only reliable comparisons.

4. Out-of-sample results (disjoint 48-text holdout)

Lane	Gateway	n	AUC	wrong	deferred	acc/corpus	decisive miss
kimi-k3 (run 1)	NVIDIA free	48	1.000	0	2	0.958	none
kimi-k3 (run 2, same night)	NVIDIA free	48	1.000	0	3	0.938	none
claude-sonnet-4.5	router/OAuth	48	—	1	0	0.979	hu-007 → AI (p 0.72)
glm-5.3	NVIDIA free	48	—	1	2	0.938	ai-012 → HUMAN (p 0.35)

- **In-sample perfection transferred for one lane and failed for another.** `glm-5.3` had erred zero times in sample with every AI text above 0.72; on the holdout its AI floor collapsed to 0.35 — the celebrated 0-wrong property was a fact about those 48 texts. `kimi-k3` kept zero wrong on the holdout *and* reproduced it in an independent same-night rerun.
- **The recurring adversarial text.** One holdout AI text (`ai-012`) defeated every other configuration that met it (glm-5.3: 0.35; the paid ensemble on cleaned text: an AI→HUMAN miss at 0.38). kimi-k3 decided it correctly as AI in both runs (0.78 / 0.74, confidence ≥ 0.6 both). One text is not proof, but it is the second occurrence of the same failure class — false negatives concentrate, and measuring them needs a two-sided holdout, not a bigger pile of human text.
- **Jitter direction on kimi-k3 was benign.** Between the two identical-parameter runs, 33 of 48 scores drifted (±0.04–0.10 typical) and exactly 3 verdicts moved — one defer became a correct AI call, two correct HUMAN calls became defers — *none moved into a wrong decision*. On this lane, jitter buys review volume, not error.
- **Pairs differ by too little to claim.** McNemar exact tests on all 48 shared texts: kimi vs sonnet, one discordant pair (kimi right); kimi vs glm-5.3, one discordant pair (kimi right); both p = 1.0000. At n = 48 a one-error advantage is not statistically separable — the defensible claim is structural (the only zero-wrong-on-both-corpora, reproduced), not statistical dominance.

5. Gateway regimes: availability is per-provider, dated, and decisive

5.1 OpenRouter free pool — two 429 classes with opposite remedies

- *Class (a), upstream per-model shared pool*: transient, can last indefinitely, **unfixable by adding accounts** — one model returned 429 on all 6 accounts × all retries in three independent confirmations spanning hours (0/12, 72 requests, nothing recovered).
- *Class (b), per-account daily cap*: printed inside the 429 body (visible only on failure; successful responses carry no rate headers); **stacks linearly across accounts** — a 6-account rotation recovered 32 missing capture rows across two quota windows with zero harness-side failures. The visible cap number varies by account history (50/day measured on no-credit accounts, up to 200/day where credits exist).
- A **404 is terminal**: once a free id is delisted its lane freezes permanently at whatever coverage it had; AUC must always be quoted with *n*.
- Gateway errors also arrive *inside* HTTP 200 bodies; harnesses that only retry on transport status misattribute these as model failures and understate the model.

5.2 Google AI Studio free tier — per-model buckets whose delivered volume is per-lane and per-project

- Quota metric is *per project, per model, per day*, stated at 20 requests; actual **delivered** volume varied enormously by lane and project: preview-family lanes delivered ~5–6 rows/day on project 1 and 16 rows in one window on project 2, while a lite-family lane delivered **46 of 48 rows in a single window**. Sleeping past the server's own `retryDelay` bought exactly one row, not a bucket — the printed delay is not a refill signal. Preview model names bill into the base model's bucket; `-latest` aliases into another.
- Because the bucket is per project, **additional project keys stack capacity linearly** — the same property OpenRouter's account rotation shows — and each key must still be probed per model: a third key that Google rejects with `400 Invalid Auth key` contributes nothing.
- Two independent limiters: a per-model bucket can be exhausted while the model still returns "high demand" 503s (these *are* transient), so capacity noise and quota starvation coexist.
- Consequence for throughput planning: **delivered requests/day/model is a measured per-lane fact, not a gateway constant** — any production detector on this tier must quota-probe its own exact model on its own project before sizing. On this gateway quota, not accuracy, was the binding constraint even though its best screen (a flash-family model) was the strongest measured anywhere: AUC 1.000, 0 wrong, human median 0.06 on 12 texts; its 21-row partial capture at window close still reported AUC 1.000 with 0 wrong.
- Consequence for validation, from the high-throughput lite lane: it delivered a *full corpus in one window* and was still band-dead — AUC 0.955 with **13 wrong calls, every one a human text read as AI** (human class bimodal, median 0.835, upper mode overlapping

the AI class). High free throughput is a reason to *measure* a lane faster, never evidence that the lane is a detector.

5.3 NVIDIA NIM free tier — serving eras, not service levels

- The catalogue endpoint lists 80 models; 55 return "function not found" on real calls. */v1/models is a promise list, not an entitlement list*.
- Models *revive and die in hours*: one opened a ~12-hour serving era (12/12 screen, AUC 1.000, the lowest human median ever measured at 0.075), collapsed to 2/48 overnight and answered nothing the next morning — an era boundary, not a time-of-day effect. Probe before spending a capture; a 2-text gate costs almost nothing.
- Latency is pool load, never a model property: p50 for the same lane moved 13.6 s → 96 s between windows on the same day; a full corpus took 144 s once and 5183 s another time, both with zero failures.
- Hard parameter constraints exist and are discoverable only in the 400 body (e.g. one model fixes top_p at 0.95 and rejects any override). *A validation 400 is a harness constraint, not a model verdict*.

5.4 Local multi-account router

- Aggregating many upstream accounts (OAuth and keys) behind one OpenAI-compatible endpoint proved the **most reliable capture channel measured**: two full-corpus runs, 48/48 both, zero salvage/truncation/errors.
- Its virtual "auto" routes rank well but band badly (§3) — a router's routing objective is not a detector's decision objective.
- Free OAuth tiers behind such routers carry provider-side risk (session expiry, policy change) that a direct key does not; treat them as convenience channels, not SLOs.

6. Which free model to use today (decision table)

Conditions of use: October 2026, short-form English prose (~85–200 words), a strict-JSON single-prompt protocol, and the band in §2.3. Re-measure before deploying: every row here is a dated measurement, not a subscription.

Rank	Lane	Gateway	acc/corpus in-sample	acc/corpus holdout	approx. median latency (s)	main constraint
1	kimi-k3	NVIDIA free	0.938	0.958 (reproduced)	14	top_p must be 0.95; serving eras can end
2	claude-sonnet-	router	0.958	0.979	3	OAuth session, not a key; 2

	4.5 via OAuth router					wrong (both human text)
3	glm-5.3	NVIDIA free	0.938	0.938	22	one AI→HUMAN miss on holdout; pool intermittently silent
4	ling-3.1-flash	OpenRouter free	0.729	not yet measured	—	12/48 deferred; best free-OR rank (AUC 0.986)
5	glm-5.3-flash	NVIDIA free	0.938	not yet measured	8	fastest strong lane; holdout pending
—	Google flash-family (3.8-flash)	AI Studio free	partial: 21/48, AUC 1.000, 0 wrong	—	18	delivered volume is per model *and* per project; stack project keys, quota-probe before sizing

- **kimi-k3's holdout row above is its first run.** The same-night rerun also had 0 wrong but deferred 3 texts instead of 2 (acc/corpus 0.938): what reproduces exactly is the absence of errors; the deferral count is what jitter moves.
- No free lane here has been verified end-to-end through a live webhook at time of writing; the paid two-model pair is the only configuration ever checked through production plumbing, and its live score (41/48) was *below* its offline replay (45/48) — a structural gap, because a secondary calibrated on one primary run is untested against a drifting primary run. Deploy-time evidence must be collected through the real stack.
- Ranking two lanes is free-tier-fragile by nature. A resilient low-cost setup runs the best lane as primary with a *different gateway*'s second-best as fallback, because cross-provider failure modes are independent while same-provider ones correlate.

7. Limitations

- **Small, single-source corpora.** 48 + 48 texts, both from HC3. The AI class is ChatGPT-3.5-era; transfer to modern frontier-model output is untested here and is the biggest open risk.
- **Verdicts are style proxies, not truth.** An independent commercial detector (GPTZero) run on the same corpora flagged de-formatted *fresh* AI text at 100% while reading genuine ChatGPT-era HC3 answers as 96% human — both instruments key on surface style and

era-calibrated signal, not on origin. Absolute verdicts on out-of-era text from any detector here, LLM-based included, should be treated as probabilistic.

- **Length confound.** AI texts average ~2× the words of human texts in both corpora; models may exploit length rather than style. A length-matched rerun is designed but not yet executed. Likewise the band was fitted on ~85-word text; its behaviour on 500-word documents is unmeasured.
- **Band transfer risk.** The 0.69/0.40 cuts were fitted in sample; several lanes have measurably different optimal cuts, and applying one band across models biases weaker-but-calibrated rows. Per-model bands were stored for a subset; comparisons in §4 deliberately used one rule for both sides.
- **All availability, quota and latency claims are dated.** Free tiers change without notice; the catalogue itself lies in both directions (listed models dead; unlisted weights served). Every number here says which gateway and which window it came from, and none of it should be quoted as a property of a model alone.
- **Not peer reviewed.** Single-lab (single-engineer) measurements, no held-out implementation verification by a third party.

8. Reproducibility and data availability

- Corpora are rebuildable from the public HC3 dataset with the recorded offset windows [1]; per-document provenance (url, retrieved date, char offsets, sha256) is kept for every row. Raw text is not redistributed.
- Every capture is stored as append-only per-row JSONL (id, label, p, confidence, signals, latency, error class, attempt) with a run manifest recording every parameter of the command that produced it. The scoring and paired-test code (band scorer, Mann-Whitney, exact McNemar, corpus disjointness verifier with negative control) are plain JS, <200 lines each.
- The prompt, parser rules and band constants are exactly those listed in §2; a per-model reproduction checklist is available on request to the submitting author.

References

1. [1] Cheng et al., "How Close is ChatGPT to Human Experts? Comparison Corpus, Evaluation and Detection," arXiv:2301.07597 (2022) — the HC3 corpus (six domains of real user answers paired with ChatGPT-3.5 answers).
2. [2] "A Semantic-Invariant Human ChatGPT Comparison Corpus and Its Website," arXiv:2309.02731 (2023) — extended HC3 release used as the datasets server source for row-level provenance checks.
3. [3] McNemar, Q., "Note on the Sampling Error of the Difference Between Correlated Proportions or Percentages," Psychological Methods, vol. 2, 1942. (Exact two-sided binomial form used here.)

4. [4] Mann, H.B., Whitney, D.R., "On a Test of Whether one of Two Random Variables is Stochastically Larger than the Other," Annals of Mathematical Statistics, 1947. (AUC computation: ties at 0.5.)